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MICA: Missing Information Comparison & Analysis Webapp

AIM

Create an interactive webapp to impute, visualise and compare data imputation techniques, helping users select the most effective strategy for data integrity.

MOTIVATION

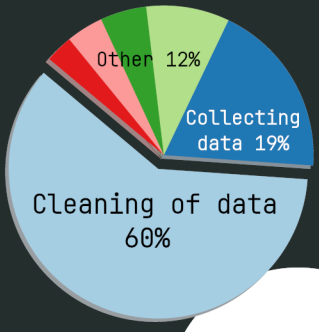
Roughly *60%* of a data scientists time is devoted to the cleaning of data. Thats the majority of the time spent!

For current implementations:
- The userflow is *manual*, *time-consuming* and *prone to error*.

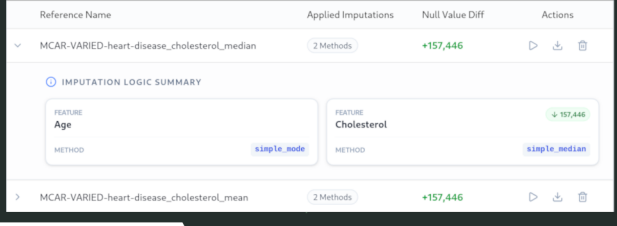
- Do not handle multiple versions of an imputation and therefore *do not have visualisations for comparing imputations*.

- Lack newly researched missing data visualisation methods.

Time Spent by Data Scientists[1, p. 3, fig: 2]



File version control UI



SOLUTION

An all in one missing data analyser, imputer and visualiser webapp with a focus on effective visualisations and usability.

MICA will:
- Create versions of a dataset using varying imputation strategies. *(implemented)*

- Keep track of imputation results via a simple version control system. *(partially implemented)*

- Provide touch screen interactive visualisations to compare results. *(not implemented)*

- Include newly developed missing data visualisation methods such as *MISSIG*. *(implemented)*

MISSIG[2]

MISSIG is a missing value visualisation method developed in 2022.

- The red box signifies the selected feature. Currently it is "ST Depression".

- The blue histograms show the distribution of the recorded feature.

- The blue bar represents the relative amount of missingness of the feature.

- The red bar is the count of missing variable that are also missing in the selected variable (red box).

- The red histogram shows the frequency of missing records where the selected variable is missing.

- In this case we can see that all the missing variables occur when age is above some threshhold.

FUTURE WORK

- LLM intergration for imputation strategy suggestions and graph analysis.

- Plugin system and uniform API for high customizability allowing users to include their own visualisation techniques.

- User studies on the effectiveness of a version control system when managing multiple versions of an imputation.

[1, p. 3, fig: 2]
M. Sarth, A. P. Tohanganj, K. Medjaher, and E. Pere, 'Data preparation and preprocessing for broadcast systems monitoring in PHM framework', in 2019 6th International Conference on Control, Decision and Information Technologies (CoDIT), Paris, France: IEEE, Apr. 2019, pp. 1444–1449. doi: 10.1109/CoDIT.2019.8820378.

[2]
S. J. Fernstad and J. J. Westberg, 'To Explore What Isn't There—Glyph-Based Visualization For Analysis of Missing Values', IEEE Trans. Visual. Comput. Graphics, vol. 28, no. 10, pp. 3514–3526, Oct. 2022. doi: 10.1109/TVCG.2021.3066124.